

Tasks for Ye Xujun

Two steps should be taken. The first step is to build an editor to help construct both informal and hybrid specifications. The second step is to develop novel methods for preventing requirements-related faults through constructing hybrid specifications.

1. Write an editor to support the construction of Informal Specifications and Hybrid Specifications.

An informal specification must conform to the style described in my book titled [Agile-SOFL: Agile Formal Engineering Method, Springer, 2024](#). The informal specification is the foundation for creating a hybrid specification that refines the informal specification.

A hybrid specification must conform to the defined grammar of Agile-SOFL. A hybrid specification is characterized by the fact that all non-bottom level processes in modules are specified using semi-formal FSF specifications and all bottom level processes are specified using formal FSF specifications. All functions are defined using either semi-formal or formal FSF specifications.

2. Propose and develop a novel method for preventing requirements-related faults through constructing hybrid specifications from the corresponding informal specification.

The essential idea is that when hybrid specifications are written, it will force the developer to think about the details in relation to requirements. As a result, all the ambiguities can be removed, and unnecessary faults can be prevented from going into the final code. But how to do this specifically is your research task.